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PROJECT MANAGEMENT WORKBOOK

Construction Worker Skill Passport with Project Portfolio

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Professor:	Dr. Dasari Shanti Ratnam
Organization:	SkillBridge Technologies Pvt Ltd.
Project Duration:	February 15, 2026 – May 15, 2026 (89 Calendar Days)
Total Budget:	₹12,50,000 (Twelve Lakhs Fifty Thousand Rupees)

PROJECT DESCRIPTION

The Construction Worker Skill Passport with Project Portfolio is a mobile-based platform designed to transform how skilled construction workers in India document, verify, and present their professional experience. India's construction sector employs over 51 million informal workers — including masons, plumbers, electricians, carpenters, and tile-layers — yet none of these workers have access to a formal, portable, and verifiable record of their skills or completed projects. Hiring decisions are dominated by contractor favoritism and word-of-mouth networks, entirely bypassing demonstrated competency.

Project Objectives

- Develop a mobile-first Android application enabling construction workers to build verified digital portfolios through before-and-after photo documentation of completed work.
- Implement a QR code-based supervisor verification system that allows site supervisors to authenticate worker contributions in real time, lending credibility to each portfolio entry.
- Integrate AI-powered image recognition to automatically tag skills such as bricklaying, plumbing, electrical wiring, tiling, and structural work from uploaded project photographs.
- Deploy a geolocation-based contractor search module enabling contractors to discover verified, skilled workers in their geographic vicinity.
- Establish a portable, contractor-agnostic worker profile that remains with the worker across employers and project sites throughout their career.
- Enable a transparent, rating-based ecosystem where workers build credibility over time through verified project outcomes and supervisor endorsements.
- Ensure platform accessibility through image compression, offline data queuing, and a lightweight app footprint suitable for 2G/3G mobile environments prevalent at Indian construction sites.

The Construction Worker Skill Passport will reduce informal hiring practices, improve labour market transparency, and empower individual workers with verifiable professional credentials — creating a merit-based employment ecosystem for India's construction sector.

PART B — INITIATING PROCESS GROUP

The Initiating Process Group formally authorizes the project and identifies the stakeholders who will influence or be impacted by the project. This group encompasses two key processes:

- Process 4.1 — Develop Project Charter
- Process 13.1 — Identify Stakeholders

Project Charter

Field	Details
Project Title	CONSTRUCTION WORKER SKILL PASSPORT WITH PROJECT PORTFOLIO
Organization	SkillBridge Technologies Pvt. Ltd.
Start Date	February 15, 2026
End Date	May 15, 2026
Total Duration	89 Calendar Days | Critical Path Sequential Sum: 89 Days
Total Effort	289 Person-Days
Project Champion	Shailaja Kanukuntla, Project Manager
Description & Justification	India's construction sector employs over 51 million informal workers who cannot formally prove their skills or work history. Hiring is dominated by contractor networks and favoritism. 1. No Formal Skill Record: Workers complete hundreds of projects with no portable, verifiable proof of competency. 2. Contractor Favoritism: Hiring decisions ignore demonstrated skill and rely entirely on personal referrals. 3. Limited Worker Mobility: Without transferable credentials, workers cannot freely approach new contractors. 4. Suppressed Wages: Skilled and unskilled workers are often compensated identically due to the absence of verifiable differentiation.

High-Level Requirements	• Worker Visual Portfolio Module • QR-Based Supervisor Verification • AI-Powered Skill Tagging (10 categories) • Geolocation-Based Contractor Search • Worker Rating & Endorsement System • Offline-Capable Low-Bandwidth Android App (≤ 25 MB APK)
Success Criteria & Measurement Owner	1. Minimum 200 active verified worker profiles onboarded within 60 days of pilot launch. [Measured by: Project Manager] 2. At least 85% of supervisor verifications completed within 24 hours of submission. [Measured by: QA Lead] 3. Platform uptime ≥ 99.5% across the first 90 days post-deployment. [Measured by: DevOps Engineer] 4. AI skill tagging accuracy ≥ 80% validated through manual QA audit. [Measured by: AI/ML Engineer] 5. Minimum 50 registered contractor accounts using geolocation search within 3 months of launch. [Measured by: Business Development Lead]
Project Budget	₹12,50,000 (Twelve Lakhs Fifty Thousand Rupees)
Project Manager — Designation, Authority & Responsibility	Shailaja Kanukuntla has been appointed as Project Manager and reports to the CEO of SkillBridge Technologies. The Project Manager holds full authority over project budget allocation, team deployment, vendor engagement, scope decisions, and milestone approvals.
Assumptions	1. Android smartphones (minimum Android 6.0) are available to or accessible by pilot workers. 2. Construction site supervisors will be formally onboarded as verified users before the pilot. 3. AWS India region will be used for all data storage to comply with data localization norms.
Constraints	1. Project must be completed within 89 calendar days. 2. Budget is fixed at ₹12,50,000. 3. Pilot launch limited to one city (Pune)

Stakeholder Identification and Ranking

Stakeholder List

Stakeholder	Role	Engagement Strategy
Shailaja Kanukuntla	Project Manager / CEO	Manage Closely
Arjun Sharma	Head of Engineering	Keep Satisfied
Priya Nair	UI/UX & Design Lead	Keep Informed
Rahul Mehta	QA & Testing Lead	Keep Informed
Vikram Joshi	AI/ML Engineer	Keep Informed
Kiran Desai	DevOps Engineer	Monitor
Sunita Rao	Field Coordinator	Monitor
Site Supervisors	Field Verification Partners	Keep Informed
Construction Workers	Primary End Users	Keep Informed
Contractor Firms	Secondary Platform Users	Keep Informed

How to rank the stakeholders?

Use the following approach to determine the rank of participating parties. Identify the parties involved in the outcome of the project and have considerable power to help achieve the project goals.

a. **High power and high interest**

Outcome: **Shailaja Kanukuntla (Project Manager)** – Holds full authority over the project and has strong interest in the successful implementation of the Construction Worker Skill Passport platform.

b. **High power but low interest**

Outcome: **Head of Engineering** – Holds technical decision-making authority regarding system architecture and development but focuses mainly on overseeing engineering operations rather than daily project activities.

c. **Medium power but high interest**

Outcome: **UI/UX Design Lead** – Plays a key role in designing the application interface and ensuring usability for construction workers, with strong interest in the success of the system.

d. **Medium power and medium interest**

Outcome: **QA((Quality Assurance) & Testing Lead** – Responsible for testing the platform and ensuring system functionality and reliability, with moderate influence and interest in project outcomes.

Stakeholder Register

Stakeholder	Rank	Role	Goal / Expected Outcome
Shailaja Kanukuntla	High Power, High Interest	Project Manager	Leads the venture; drives strategic decisions for commercial success and worker empowerment.
Arjun Sharma	High Power, Low Interest	Head of Engineering	Oversees technical architecture; ensures delivery meets performance and scalability benchmarks.
Priya Nair	Medium Power, High Interest	UI/UX & Design Lead	Designs accessible interfaces for low-literacy construction workers to ensure adoption.
Rahul Mehta	Medium Power, Medium Interest	QA & Testing Lead	Validates system reliability, QR accuracy, and AI tagging performance.
Vikram Joshi	Medium Power, High Interest	AI/ML Engineer	Builds and maintains skill tagging AI model with ≥ 80% accuracy.
Kiran Desai	Low Power, Low Interest	DevOps Engineer	Manages cloud deployment, CI/CD pipelines, and system uptime.

Stakeholder	Rank	Role	Goal / Expected Outcome
Sunita Rao	Low Power, Medium Interest	Field Coordinator	Facilitates UAT sessions at construction sites; supports pilot worker onboarding.
Site Supervisors	Low Power, High Interest	Field Verification Partners	Authenticate worker contributions via QR codes; critical to portfolio credibility.
Construction Workers	Low Power, Very High Interest	Primary End Users	Build verified portfolios; gain portable, merit-based professional credentials.
Contractor Firms	Medium Power, High Interest	Secondary Platform Users	Discover and hire verified workers; reduce hiring risk and improve labour quality.

PLANNING PROCESS GROUP

The Planning Process Group comprises processes required to establish the total scope of the project, define and refine objectives, and develop the course of action required to attain those objectives. According to PMBOK 6th Edition, this group produces the Project Management Plan and all subsidiary plans.

Project Management Plan

Project Name	Construction Worker Skill Passport with Project Portfolio
Prepared By	Shailaja Kanukuntla, Project Manager
Date	February 15, 2026
Scope Statement	• Android mobile application development for three user roles: Worker, Supervisor, and Contractor. • Backend API and cloud infrastructure (AWS) for portfolio storage, verification, and matching. • QR code-based supervisor verification system integrated into the Android app. • AI-powered image recognition for automated skill tagging of construction work photographs. • Geolocation-based contractor search and worker discovery module. • Offline-first architecture with automatic sync for low-connectivity field environments. • Pilot deployment with 200 verified workers and 50 contractors in Pune, Maharashtra.

Subsidiary Plans Overview

Subsidiary Plan	Brief Description
Scope Management Plan	Defines how project scope is defined, validated, and controlled throughout the lifecycle.
Schedule Management Plan	Defines scheduling methodology, tools (Gantt / Critical Path), and update frequency.

Subsidiary Plan	Brief Description
Cost Management Plan	Defines budgeting method, cost control thresholds, and reporting format.
Quality Management Plan	Defines quality standards, metrics, assurance activities, and control procedures.
Resource Management Plan	Defines team roles, acquisition strategy, RACI assignments, and staff release procedures.
Communications Management Plan	Defines message types, frequency, methods, senders, and recipients.
Risk Management Plan	Defines risk identification, scoring, response strategies, and contingency planning.
Procurement Management Plan	Defines vendor sourcing, evaluation, contracting approach, and procurement timeline.
Change Management Plan	Defines the Change Request process, approval authority, and change log maintenance.

Requirements Collection and Scope Statement

Requirement Register

Category	Requirement	Stakeholder	Acceptance Criteria
Funds	₹12,50,000 for Android development, AI model training, cloud infrastructure, testing, and pilot deployment	Project Manager, Sponsors	100% budget secured prior to project start
Design	Android UI/UX with icon-based navigation for low-literacy workers; visual portfolio gallery; supervisor QR scan interface; contractor search map view	UI/UX Lead, Workers, Contractors	UI prototype approved with usability score ≥ 4/5
Operations	Platform accessible to workers in Tier-2 and Tier-3 cities with 2G/3G connectivity; offline queuing for photo uploads	Project Manager, Workers	Offline mode tested and validated in UAT(User Acceptance Testing)
Technology	AI skill tagging accuracy ≥ 80%; QR verification turnaround ≤ 24 hours; app APK size ≤ 25 MB; API response time ≤ 2 seconds	AI/ML Engineer, QA Lead	All technical benchmarks validated in load testing 1.AI tagging accuracy ≥ 80% 2.API response ≤ 2 seconds 3.APK size ≤ 25 MB
Security	Worker data encrypted at rest (AES-256) and in transit (TLS); OTP-based authentication; role-based access control; DPDP Act 2023 compliance	Backend Engineer, Compliance Officer	Zero critical vulnerabilities in penetration test report 0 vulnerabilities → Accepted ≥1 vulnerability → Rejected

Project Scope Statement

This project aims to build a mobile-based credential and portfolio system for informal construction workers across India. The platform will use visual work documentation, QR-based peer verification, and AI-powered skill tagging to create portable, contractor-agnostic worker profiles — enabling merit-based hiring and reducing contractor favoritism in the Indian construction labour market.

Project Objective

To develop a scalable, low-bandwidth Android application that enables construction workers to build verified digital skill portfolios, connects them with contractors through geolocation matching, and creates a transparent, skill-based labour marketplace for India's construction sector — within an 89-day timeline and ₹12,50,000 budget.

Deliverables

- Fully functional Android application with Worker, Supervisor, and Contractor modules
- QR code verification system integrated with supervisor approval workflows
- AI-powered skill tagging engine validated at minimum 80% accuracy
- Cloud-hosted backend (AWS) with portfolio storage and geolocation search capability
- Pilot deployment report: 200 verified workers and 50 contractors onboarded in Pune
- Post-launch maintenance and monitoring plan

Limits and Exclusions

- iOS development is excluded from version 1.0 scope
- AI skill tagging limited to 10 predefined skill categories in initial release
- Pilot launch restricted to Pune, Maharashtra; multi-city rollout is out of scope for this phase
- Background verification or police clearance certificate integration is excluded from current scope

Acceptance Criteria

The system must onboard 200 verified worker profiles and 50 contractor accounts within 60 days of pilot launch. AI tagging accuracy must exceed 80% in independent QA review. Platform uptime must remain above 99.5% throughout the pilot period. Worker satisfaction (post-onboarding survey) must reach or exceed 80% positive.

Signatures:

[Shailaja Kanukuntla], Project Manager — SkillBridge Technologies Pvt. Ltd.

Work Breakdown Structure (WBS)

The WBS decomposes the total project scope into manageable work packages. It is organized hierarchically, with the project at Level 1 (root node), major deliverable groups at Level 2, and individual work packages at Level 3. All work required to complete the project is represented within the WBS, in compliance with PMBOK 6th Edition, Section 5.4.

Project: Construction Worker Skill Passport — WBS Hierarchy

1.0 Construction Worker Skill Passport Platform

1.1 Project Initiation & Planning

- 1.1.1 Problem Research & Field Interviews with Workers
- 1.1.2 Feasibility Study & Competitor Analysis
- 1.1.3 Stakeholder Identification & Power-Interest Mapping
- 1.1.4 Project Charter, Budget, and Timeline Finalization

1.2 Platform Design & Architecture

- 1.2.1 User Journey Mapping (Worker, Supervisor, Contractor Flows)
- 1.2.2 UI/UX Wireframes & Clickable Figma Prototype
- 1.2.3 Backend Microservices Architecture Design
- 1.2.4 Database Schema Design & REST API Specification

1.3 Core Application Development

- 1.3.1 Android App — Worker Portfolio & Photo Upload Module
- 1.3.2 Android App — Supervisor QR Verification Module
- 1.3.3 Android App — Contractor Search & Discovery Module
- 1.3.4 Backend API Development & AWS Cloud Storage Integration
- 1.3.5 Worker Rating & Endorsement System

1.4 AI & Geolocation Feature Development

- 1.4.1 Construction Image Training Data Collection & Annotation
- 1.4.2 AI Skill Tagging Model Training & Optimization (MobileNet)
- 1.4.3 AI Model Backend Integration & API Endpoint
- 1.4.4 Geolocation Search Engine Integration (Google Maps API)
- 1.4.5 Image Compression Module & Offline Queue Development

1.5 Testing & Quality Assurance

- 1.5.1 Unit Testing — All Android Modules
- 1.5.2 Integration Testing — Backend API & Cloud Services

1.5.3 QR Verification Accuracy & End-to-End Flow Testing

1.5.4 AI Tagging Performance Validation (500-image test set)

1.5.5 Security Penetration Testing

1.5.6 User Acceptance Testing (UAT) with Construction Site Workers

1.6 Deployment & Pilot Launch

1.6.1 AWS Cloud Infrastructure Provisioning & CI/CD Pipeline Setup

1.6.2 Google Play Store App Publication

1.6.3 Pilot Worker & Contractor Onboarding (Pune)

1.6.4 Field Monitoring, Issue Resolution & Support

1.6.5 Post-Pilot Report & Maintenance Handover Documentation

Schedule and Milestones

Project Start: February 15, 2025 | Project End: May 15, 2025 | Calendar Span: 89 Days | Critical Path Sequential Sum: 90 Days

WBS Activity	Days	Start	End	Float	Milestone / Deliverable
1.1 Project Initiation & Planning	14	Feb 15	Feb 28	0	Project Charter approved; Stakeholder register finalized; Budget confirmed
1.2 Platform Design & Architecture	14	Mar 1	Mar 14	0	UI/UX wireframes approved; Backend architecture signed off; DB schema finalized
1.3 Core Application Development	21	Mar 15	Apr 4	0 ★CP	Android app modules (Worker, Supervisor, Contractor) developed; Backend API functional
1.4 AI & Geolocation Development	14	Mar 15	Mar 28	+7 Float	AI model trained; Geolocation module integrated; Image compression deployed (parallel with 1.3)
1.5 Testing & Quality Assurance	14	Apr 5	Apr 18	0	All modules tested; Security pen test passed; UAT completed with construction workers
1.6 Deployment & Pilot Launch	27	Apr 19	May 15	0	App live on Play Store; 200 workers and 50 contractors onboarded; Pilot report submitted

★ CP = Critical Path Activity | Sequential phase sum (critical path): 14+14+21+14+27 = 90 days | Activity 1.4 runs parallel to 1.3, saving 7 days, giving a calendar span of 89 days (Feb 15 – May 15) ✓

Milestone List

Milestone	Target Date
M1 — Project Charter Approved & Team Fully Onboarded	February 28, 2025

Milestone	Target Date
M2 — UI/UX Prototype & Backend Architecture Finalized	March 14, 2025
M3 — AI Skill Tagging Model Training Complete	March 28, 2025
M4 — Core Android App Modules Development Complete	April 4, 2025
M5 — All Testing & UAT Complete; Security Pen Test Passed	April 18, 2025
M6 — App Published on Google Play Store	April 25, 2025
M7 — Pilot: 200 Workers + 50 Contractors Onboarded	May 10, 2025
M8 — Full Project Completion & Post-Pilot Report Submitted	May 15, 2025

Activity Dependency (Network Logic)

Activity	Dependency Type	Predecessor
1.1 Project Initiation	—	None (Start of project)
1.2 Platform Design	Finish-to-Start (FS)	Must follow 1.1 completion
1.3 Core Development	Finish-to-Start (FS)	Must follow 1.2 completion
1.4 AI & Geolocation Dev	Start-to-Start (SS) with 1.3	Begins on same day as 1.3; parallel track with 7 days float
1.5 Testing & QA	Finish-to-Start (FS)	Must follow 1.3 AND 1.4 completion
1.6 Deployment & Pilot	Finish-to-Start (FS)	Must follow 1.5 completion

Critical Path Analysis

The critical path is the longest sequence of dependent activities that determines the minimum project duration. Float = 0 on the critical path.

Path	Activities	Duration	Status
Critical Path	1.1 → 1.2 → 1.3 → 1.5 → 1.6	14+14+21+14+27 = 90 days	★ CRITICAL

Path	Activities	Duration	Status
Non-Critical (Parallel)	1.4 AI & Geolocation Dev	14 days	7 Days Float

Activity Duration and Resource Estimation

Note on Person-Day Calculation: Person-Days = Activity Duration (Days) × Number of Resources Assigned. UAT participant workers (end users) are not counted as project staff resources.

Activity	Days	Person-Days	Resources Assigned
1.1.1 Problem Research & Field Interviews	5	10	Project Manager (1), Business Analyst (1)
1.1.2 Feasibility Study & Competitor Analysis	5	5	Project Manager (1)
1.1.3 Stakeholder Mapping	2	2	Project Manager (1)
1.1.4 Charter, Budget & Timeline Finalization	2	4	Project Manager (1), Finance Lead (1)
1.2.1 User Journey Mapping	3	6	UI/UX Designer (1), Design Lead (1)
1.2.2 UI/UX Wireframes & Prototype	7	14	UI/UX Designer (1), Design Lead (1)
1.2.3 Backend Architecture Design	5	10	Head of Engineering (1), Backend Engineer (1)
1.2.4 Database Schema & API Specification	4	8	Backend Engineer (1), Head of Engineering (1)
1.3.1 Android — Worker Portfolio Module	8	16	Senior Android Developer (1), Junior Android Dev (1)
1.3.2 Android — Supervisor QR Module	6	12	Senior Android Developer (1), Backend Engineer (1)
1.3.3 Android — Contractor Module	5	10	Junior Android Developer (1), Backend Engineer (1)
1.3.4 Backend API & Cloud Integration	8	16	Backend Engineer (1), DevOps Engineer (1)

Activity	Days	Person-Days	Resources Assigned
1.3.5 Worker Rating System	4	8	Senior Android Developer (1), Backend Engineer (1)
1.4.1 Training Data Collection & Annotation	5	10	AI/ML Engineer (1), Field Coordinator (1)
1.4.2 AI Model Training & Optimization	5	10	AI/ML Engineer (1), Backend Engineer (1)
1.4.3 AI Model Backend Integration	2	4	AI/ML Engineer (1), Backend Engineer (1)
1.4.4 Geolocation Search Integration	3	6	Backend Engineer (1), Android Developer (1)
1.4.5 Image Compression & Offline Queue	4	8	Senior Android Developer (1), Junior Android Developer (1)
1.5.1 Unit Testing — Android Modules	4	8	QA Lead (1), QA Tester (1)
1.5.2 Integration Testing — Backend & Cloud	3	6	QA Lead (1), Backend Engineer (1)
1.5.3 QR Verification Testing	2	4	QA Lead (1), QA Tester (1)
1.5.4 AI Tagging Performance Validation	3	6	QA Lead (1), AI/ML Engineer (1)
1.5.5 Security Penetration Testing	3	6	External Security Firm (2-person team)
1.5.6 User Acceptance Testing (UAT)	5	20	QA Lead (1), Field Coordinator (1), UAT Facilitators (2)
1.6.1 Cloud Infrastructure & CI/CD Setup	3	6	DevOps Engineer (1), Backend Engineer (1)

Activity	Days	Person-Days	Resources Assigned
1.6.2 Google Play Store Publication	2	4	Senior Android Developer (1), DevOps Engineer (1)
1.6.3 Pilot Worker & Contractor Onboarding	12	36	Field Coordinator (1), Project Manager (1), Support Staff (1)
1.6.4 Field Monitoring & Issue Resolution	8	24	All Teams (on-call rotation, 3 per shift)
1.6.5 Post-Pilot Report & Maintenance Handover	5	10	Project Manager (1), QA Lead (1)
TOTAL	**90**	**289**	**Across all 29 work packages**

Summary by Phase

Phase	Duration (Days)	Total Person-Days	Peak Headcount
1.1 Project Initiation	14	21	2
1.2 Platform Design & Architecture	14	38	2
1.3 Core Application Development	21	62	3
1.4 AI & Geolocation Development	14 (parallel with 1.3)	38	2
1.5 Testing & Quality Assurance	14	50	4
1.6 Deployment & Pilot Launch	27	80	5
TOTAL	**90 days (critical path sequential sum) 89-day calendar span: Feb 15 – May 15**	**289 Person-Days**	**~9 Peak**

Deliverables List by Team

Team	Activities and Deliverables
Team 1 — Design	User journey maps (1.2.1); UI/UX wireframes and Figma prototype (1.2.2); design system and icon library
Team 2 — Android Dev	Worker portfolio module (1.3.1); Supervisor QR module (1.3.2); Contractor search module (1.3.3); Rating system (1.3.5); Image compression (1.4.5)
Team 3 — Backend & Cloud	Backend API (1.3.4); AWS S3 integration; QR token system; Geolocation API (1.4.4); AI model API endpoint (1.4.3); CI/CD pipeline (1.6.1)
Team 4 — AI/ML	Training dataset preparation (1.4.1); MobileNet model training (1.4.2); Model validation and optimization
Team 5 — QA & Testing	All test plans and execution (1.5.1–1.5.6); defect register; UAT report; penetration test coordination
Team 6 — Field & PM	Field research (1.1.1); UAT facilitation (1.5.6); pilot onboarding (1.6.3); monitoring (1.6.4); post-pilot report (1.6.5)

RISK MANAGEMENT PLAN

Risks are identified, assessed, and tracked in alignment with PMBOK 6th Edition Chapter 11. Each risk is scored using a Probability × Impact matrix. Risks scoring High (H) require active mitigation strategies and contingency plans. Risks scoring Medium (M) are monitored with defined triggers. Low (L) risks are accepted and documented.

Risk Identification

ID	Risk	Type	Consequence	Impact on Project
R1	Low digital literacy among construction workers	Negative	Poor adoption; workers unable to navigate app	Schedule & Benefit Realization
R2	Supervisor reluctance to participate in QR verification	Negative	Unverified portfolios; reduced platform credibility	Scope & Quality
R3	Poor image quality from construction site photos	Negative	AI tagging inaccuracy; false skill certifications	Quality
R4	Network connectivity failures at construction sites	Negative	Workers unable to upload; data loss risk	Schedule & Quality
R5	Data privacy breach of worker information	Negative	Legal liability; loss of user trust; regulatory action	Cost & Reputation
R6	Contractor non-adoption of platform	Negative	No hiring demand; worker value proposition collapses	Benefits Realization
R7	AI model bias toward specific construction trades	Negative	Unequal skill recognition across worker categories	Quality & Equity

ID	Risk	Type	Consequence	Impact on Project
R8	Regulatory non-compliance for labour data storage	Negative	Platform shutdown; legal penalties	Cost & Legal
R9	App performance degradation on low-end Android devices	Negative	Exclusion of target worker segment	Quality & Scope
R10	Freelancer dropout or unavailability	Negative	Schedule delay; cost overrun	Schedule & Cost

Risk Probability × Impact Matrix

Green = Low Risk | Yellow = Medium Risk | Red = High Risk

Probability / Impact →	Low Impact	Medium Impact	High Impact
High (>60%)	—	R2 — Supervisor Reluctance R10 — Freelancer Dropout	R4 — Network Failure R1 — Low Digital Literacy
Medium (30–60%)	R7 — AI Model Bias	R3 — Poor Image Quality R6 — Contractor Non-Adoption R9 — App Performance	R5 — Data Privacy Breach
Low (<30%)	R8 — Regulatory Non-Compliance	—	—

Risk Register — Quantification and Response Strategy

ID	Risk	Prob.	Impact	Response Strategy	Rating After
R1	Low Digital Literacy	High	High	Mitigate — Icon-based visual UI; voice guidance in regional languages; on-site	Medium

ID	Risk	Prob.	Impact	Response Strategy	Rating After
				onboarding workshops with field coordinator	
R2	Supervisor Reluctance	High	Medium	Prevent — Simplify QR scan to 2-tap flow; partner with contractor firms to mandate participation; offer recognition badges	Low
R3	Poor Image Quality	Medium	Medium	Enhance — In-app image quality checker with guided framing overlay; auto-reject and re-prompt for low-resolution uploads	Low
R4	Network Failure	High	High	Mitigate — Offline-first architecture; local data queue; automatic background sync on reconnect	Medium
R5	Data Privacy Breach	Medium	High	Prevent — AES-256 encryption at rest; TLS 1.3 in transit; OTP auth; quarterly pen testing; DPDP Act 2023 compliance	Medium
R6	Contractor Non-Adoption	Medium	Medium	Mitigate — Free 6-month trial tier; dedicated onboarding support; ROI case studies with pilot contractors	Low
R7	AI Model Bias	Medium	Low	Enhance — Balanced training dataset across 10 skill categories; quarterly bias audits	Low
R8	Regulatory Non-Compliance	Low	Low	Accept — Legal review pre-launch; DPDP Act 2023 compliance checklist implemented	Low
R9	App Performance	Medium	Medium	Mitigate — APK ≤ 25 MB target; image compression; Firebase Test Lab testing on low-end devices	Low
R10	Freelancer Dropout	High	Medium	Mitigate — Milestone contracts with 20% retention bonus; backup candidates pre-identified for critical roles	Low

Risk Contingency Plan

ID	Risk	Trigger Event	Contingency Steps
R1	Low Digital Literacy	Worker onboarding drop-off rate > 30% in first 2 weeks	Deploy field support agents at pilot sites; introduce WhatsApp-based photo submission as fallback registration
R2	Supervisor Reluctance	QR verification rate < 60% of submitted work entries	Escalate to contractor management; deploy SMS-based backup verification; offer per-verification incentive
R4	Network Failure	Upload failure rate > 20% at pilot sites	Pre-position offline data packs; implement SMS-based photo metadata fallback; notify workers of pending sync
R5	Data Privacy Breach	Security alert triggered or unauthorized access detected	Invoke incident response plan; notify affected users within 72 hours; engage cybersecurity firm; restrict data access temporarily
R10	Freelancer Dropout	Key developer unavailable for > 5 consecutive working days	Activate pre-identified backup candidate; escalate to recruitment agency; adjust sprint scope temporarily

COST MANAGEMENT PLAN

The Cost Management Plan defines how project costs will be planned, estimated, budgeted, managed, and controlled throughout the project lifecycle. This plan is a subsidiary document of the Project Management Plan, aligned with PMBOK 6th Edition, Section 7.1.

Cost Estimation Method	Bottom-up estimation: costs estimated at work package level and aggregated upward through the WBS. Three-point estimation used for uncertain items.
Cost Baseline	₹12,50,000 — established from WBS-level cost aggregation. This serves as the authorized budget against which actual performance is measured.
Cost Control Threshold	Variance of ±10% (₹1,25,000) triggers a formal Cost Review Meeting. Variance > 15% triggers escalation to sponsor for re-baselining.
Earned Value Reporting	Monthly EVM reports tracking Cost Performance Index (CPI) and Schedule Performance Index (SPI). Target CPI ≥ 1.0; SPI ≥ 0.95.
Contingency Reserve	₹62,500 (5% of budget) held as contingency reserve for identified risks. Management reserve of ₹25,000 held by sponsor for unknown unknowns.
Cost Reporting Frequency	Monthly financial reports to CEO. Biweekly cloud cost monitoring. Per-milestone vendor payment reconciliation.
Budget Update Authority	Project Manager may reallocate up to ₹50,000 within approved categories without sponsor approval. Changes above ₹50,000 require formal Change Request.

Cost Estimation

Cost estimation follows bottom-up methodology, estimating costs at the work package level and aggregating upward through the WBS. All figures reflect Indian market rates as of early 2025 for construction technology projects.

1. Cloud Infrastructure & Hosting

S.No	Element	Duration	Estimated Cost (INR)
1	AWS EC2 (t3.medium) — Backend API Server	3 months	₹30,000
2	AWS S3 — Portfolio Image Storage (estimated 500 GB)	3 months	₹18,000
3	AWS RDS (PostgreSQL) — Database Instance	3 months	₹15,000
4	AWS CloudFront CDN — Image Delivery	3 months	₹8,000
5	Firebase Realtime DB & Authentication (Blaze Plan)	3 months	₹12,000
6	GPU Instance (AWS EC2 p3.2xlarge) — AI Model Training	2 weeks	₹25,000
7	Google Play Store Developer Account (one-time)	—	₹2,000
8	Domain Registration + SSL Certificate (1 year)	—	₹5,000
9	CDN & DNS (Cloudflare)	3 months	₹3,000
	Total Infrastructure Cost:		**₹1,18,000**

2. Software Tools & Licenses

S.No	Material / Tool	Duration	Estimated Cost (INR)
10	Figma Professional Plan	3 months	₹6,000
11	Jira + Confluence (Atlassian Team Plan)	3 months	₹9,000
12	Postman API Testing (Team Plan)	3 months	₹5,000
13	OWASP ZAP (open-source) + security license supplement	—	₹3,000

S.No	Material / Tool	Duration	Estimated Cost (INR)
14	Firebase Test Lab Credits	—	₹7,000
15	Google Maps Platform API Credits	3 months	₹7,000
	Total Tools & Licenses Cost:		**₹37,000**

3. Manpower

S. No	Role	Count	Mo.	Monthly Rate	Calculation	Total Cost
16	Project Manager (Founder)	1	3	₹0 (Equity)	₹0 × 3	₹0
17	Senior Android Developer	1	3	₹60,000	₹60,000 × 3	₹1,80,000
18	Junior Android Developer	1	2	₹38,000	₹38,000 × 2	₹76,000
19	Backend Engineer (Node.js)	1	3	₹65,000	₹65,000 × 3	₹1,95,000
20	AI/ML Engine	1	2	₹70,000	₹70,000 × 2	₹1,40,000

S. No	Role	Count	Mo.	Monthly Rate	Calculation	Total Cost
	er (Freelance)					
21	UI/UX Designer	1	2	₹45,000	₹45,000 × 2	₹90,000
22	QA Lead	1	2	₹50,000	₹50,000 × 2	₹1,00,000
23	QA Tester (Junior)	1	1.5	₹30,000	₹30,000 × 1.5	₹45,000
24	DevOps Engineer (Contract)	1	1	₹55,000	₹55,000 × 1	₹55,000
25	Field Coordinator	1	1	₹25,000	₹25,000 × 1	₹25,000
	Total Manpower Cost:				₹0+₹1,80,000+₹76,000+₹1,95,000+₹1,40,000+₹90,000+₹1,00,000+₹45,000+₹55,000+₹25,000	**₹8,06,000**

4. Development & Operational Costs

S.No	Element	Estimated Cost (INR)
26	AI Training Dataset Procurement & Annotation (5,000 labelled images)	₹40,000

S.No	Element	Estimated Cost (INR)
27	Security Penetration Testing (External Cybersecurity Agency)	₹30,000
28	Pilot Launch Field Operations & Worker Onboarding Support	₹35,000
29	Miscellaneous (stationery, travel for field visits, contingencies)	₹20,000
30	Post-Launch Maintenance Reserve (3 months, on-call basis)	₹64,000
	Total Development & Operational Costs:	**₹1,89,000**

5. Grand Total Estimated Cost — Summary

Cost Category	Total Estimated Cost (INR)
1. Cloud Infrastructure & Hosting	₹1,18,000
2. Software Tools & Licenses	₹37,000
3. Manpower (All Roles)	₹8,06,000
4. Development & Operational Costs	₹1,89,000
GRAND TOTAL	**₹12,50,000 (~₹12.5 Lakhs)**

Budget Verification: ₹1,18,000 + ₹37,000 + ₹8,06,000 + ₹1,89,000 = ₹12,50,000 ✓

QUALITY MANAGEMENT PLAN

Project Title: Construction Worker Skill Passport | Version: 1.0 | Date: February 15, 2025

This plan ensures the Skill Passport platform delivers a reliable, secure, and user-friendly experience for construction workers, supervisors, and contractors.

Quality Roles and Responsibilities

Role	Responsibility
Project Manager	Ensures all milestone deliverables meet agreed quality criteria; approves final release
QA Lead (Rahul Mehta)	Defines test plans; manages functional, integration, security, and UAT execution; maintains defect register
AI/ML Engineer (Vikram Joshi)	Validates AI skill tagging accuracy; manages model retraining; conducts bias audits
Head of Engineering (Arjun Sharma)	Ensures API reliability, cloud uptime, and backend performance against defined SLAs
UI/UX Lead (Priya Nair)	Validates usability through field testing; ensures accessibility standards for low-literacy users
Compliance Officer	Monitors DPDP Act 2023 and IT Act 2000 compliance; coordinates penetration testing and audits

Quality Standards and Metrics

Quality Metric	Definition	Target	Measurement Method
AI Tagging Accuracy Rate	% of skill tags correctly assigned from construction photos	≥ 80%	Manual QA audit of 500-image test set
Platform Uptime	System availability across all user-facing services	≥ 99.5%	AWS CloudWatch monitoring
QR Verification Turnaround	Time from QR scan to verified badge on worker profile	≤ 24 hours	Automated timestamp logging

Quality Metric	Definition	Target	Measurement Method
Worker Satisfaction Score	Post-onboarding survey rating from pilot workers	≥ 80% positive	In-app survey form
API Response Time	Time for backend API to respond under standard load	≤ 2 seconds	JMeter load test
App Load Time (3G)	Time from launch to home screen on 3G connection	≤ 5 seconds	Firebase Test Lab
Bug Resolution Time	Time to resolve critical production bugs post-launch	≤ 48 hours	Jira SLA tracking
Test Case Pass Rate	% of test cases passing before UAT sign-off	≥ 95%	Jira test reports

Quality Reporting Plan

Report Type	Frequency	Contents
Sprint Quality Report	Biweekly	Bug counts by severity, test case pass rate, open defects, resolved defects
AI Model Accuracy Report	Monthly	Precision, recall, F1 score per skill category; bias audit results
System Health Dashboard	Continuous	API uptime, response times, error rates, cloud resource utilization
UAT Findings Report	Post-UAT (one-time)	Task completion rates, worker satisfaction scores, critical usability issues
Security Audit Report	Pre-launch + Quarterly	Penetration test findings, DPDP compliance status, remediation actions
Post-Pilot Quality Summary	End of pilot (May 15)	All KPI actuals vs. targets, lessons learned, recommendations for v1.1

HUMAN RESOURCE MANAGEMENT PLAN

Project Title: Construction Worker Skill Passport | Version: 1.0 | Date: February 15, 2025

Roles and Responsibilities

Team Member	Role	Key Responsibilities
[Your Name Here]	Project Manager / CEO	Strategic direction; budget authority; stakeholder management; milestone approvals; final sign-offs
Arjun Sharma	Head of Engineering	Backend and cloud architecture decisions; technical team management; engineering delivery oversight
Priya Nair	UI/UX & Design Lead	User research; wireframe and prototype design; design system; field usability testing oversight
Rahul Mehta	QA & Testing Lead	Test plan development; test execution management; defect tracking; UAT coordination; quality reporting
Vikram Joshi	AI/ML Engineer	Training dataset preparation; model development and training; accuracy validation; bias audits
Senior Android Developer	Android Development	Worker, Supervisor, Contractor module development; image compression; offline queue implementation
Junior Android Developer	Android Development	Support to senior developer; UI implementation; device compatibility testing
Backend Engineer	Backend Development	REST API development; AWS S3 integration; QR token system; geolocation API; database management
Kiran Desai	DevOps Engineer	CI/CD pipeline setup; AWS infrastructure provisioning; deployment; system health monitoring
Sunita Rao	Field Coordinator	Field research; UAT facilitation at construction sites; pilot worker onboarding support

RACI Matrix

R = Responsible | A = Accountable | C = Consulted | I = Informed

Activity	PM	HoE	UX	QA	AI/ML	DevOps	Field
Project Charter & Planning	A/R	C	I	I	I	I	I
UI/UX Design & Wireframes	A	I	R	C	I	I	C
Backend Architecture Design	A	A/R	C	C	C	C	I
Android App Development	A	A	C	C	I	C	I
AI Model Training & Integration	A	C	I	C	R	C	C
QA Testing & UAT	A	C	C	R	C	I	R
Cloud Deployment & CI/CD	A	A	I	C	I	R	I
Security Penetration Testing	A	C	I	R	I	C	I
Pilot Worker Onboarding	A	I	I	C	I	I	R
Post-Pilot Report	R	C	C	C	C	C	C
Budget & Cost Control	R/A	C	I	I	I	C	I
Stakeholder Communication	R/A	C	C	C	I	I	C

Staff Acquisition Plan

- Hire a Senior Android Developer with experience in offline-first architecture and TensorFlow Lite integration through Naukri or LinkedIn.
- Recruit a Backend Engineer with Node.js and AWS expertise through a developer agency for a 3-month milestone-based contract.
- Engage a Freelance AI/ML Engineer specializing in computer vision and MobileNet via Upwork or a specialized AI talent platform.
- Contract a DevOps Engineer for 1 month to set up the CI/CD pipeline and AWS infrastructure.
- Engage the Field Coordinator locally in Pune with construction sector experience for field research and UAT facilitation.

Staff Release Plan

- Freelance and contract staff are released upon deliverable acceptance and formal knowledge transfer sign-off.
- All code repositories, documentation, and credentials are transferred to the permanent team before release.
- High-performing contributors will be offered advisory or part-time maintenance roles post-launch.

Training Plan

Training Area	Details
Technical Training	Android Studio best practices; TensorFlow Lite deployment; Firebase real-time sync; AWS S3 SDK
Compliance Training	DPDP Act 2023 data handling protocols; OTP and QR security standards; data retention policies
Accessibility Training	Designing for low-literacy users; regional language UI guidelines; WCAG 2.1 basics
Agile & Collaboration	Agile sprint methodology; Jira board usage; Confluence documentation; Slack communication norms
Field Safety & Protocol	Construction site safety norms; worker interview techniques; informed consent for UAT participants

Performance Review KPIs

- Sprint delivery rate: percentage of committed story points completed per sprint (target ≥ 85%).
- Defect escape rate: bugs found in UAT that were not caught in internal testing (target < 10%).
- AI model accuracy improvement rate per training iteration.
- Worker onboarding success rate during the 27-day pilot phase.
- Freelancer milestone delivery: on-time delivery of each contracted deliverable.

COMMUNICATION MANAGEMENT PLAN

This plan defines how project information will be planned, collected, created, distributed, stored, retrieved, and reported throughout the project lifecycle, in alignment with PMBOK Process 10.1 — Plan Communications Management.

Message	Description	Report To	Method	Frequency	Sender
Daily Standup Update	Progress, blockers, and daily priorities across all development streams	Head of Engineering	Slack / Google Meet (15 min)	Daily	All Dev Leads
Sprint Progress Report	Sprint velocity, story points delivered, blockers, and next sprint plan	PM & HoE	Jira Sprint Report + Confluence	Biweekly	QA Lead / PM
AI Model Performance	Accuracy metrics, precision/recall per category, retraining status	HoE & PM	Technical Report + Confluence	Weekly	AI/ML Engineer
QA & Testing Updates	Bug severity counts, test pass rates, open defects, UAT findings	HoE & QA Lead	Jira Dashboard + Test Report	Biweekly	QA Lead
Backend & Cloud Status	API uptime, server load, storage usage, incident reports	Head of Engineering	AWS CloudWatch + Email	Weekly	DevOps Engineer
Security & Compliance	Pen test progress, DPDP compliance status, audit findings	PM & Compliance Officer	Compliance Report Email	Monthly	Compliance Officer
Budget & Resource Report	Actual vs. planned cost; headcount and cloud cost tracking	PM & Finance Lead	Google Sheets / Email	Monthly	PM / Finance

Message	Description	Report To	Method	Frequency	Sender
Pilot Launch Dashboard	Worker/contractor onboarding numbers, field issues, adoption metrics	PM & All Stakeholders	Real-time Dashboard + Briefing	Weekly (pilot)	PM
Monthly Stakeholder Briefing	Milestones, risks, decisions, and upcoming deliverables	All Stakeholders	PowerPoint Presentation	Monthly	Project Manager
Issue & Escalation Log	Documented escalations, decisions made, resolution actions	PM & HoE	Confluence Log + Email	As needed	Any Team Member

Communication Tools

Tool	Purpose
Slack	Daily team communication; channel-based (dev, qa, ai, general, pilot)
Jira	Sprint planning, task tracking, defect management, and sprint reports
Confluence	Project documentation, meeting notes, technical specs, and test plans
Google Meet / Zoom	Sprint reviews, stakeholder briefings, and remote team meetings
AWS CloudWatch	Real-time system health and infrastructure monitoring
Google Workspace	Budget tracking, documents, and stakeholder reports

PROCUREMENT MANAGEMENT PLAN

1. Procurement Objectives

- To acquire skilled technical resources, cloud services, and AI development tooling within the ₹12,50,000 approved project budget.
- To ensure timely contracting and onboarding of all freelance and contract staff before their respective sprint start dates.
- To select vendors through transparent, criteria-based evaluation covering experience, pricing, and delivery track record.
- To maintain full compliance with IT Act 2000 and DPDP Act 2023 in all vendor agreements involving worker data handling.

2. Procurement Methods

Method	Description	When Used
Direct Purchase	Procuring standard subscriptions or tools with published pricing directly from providers	AWS, Firebase, Figma, Google Play, Atlassian
Request for Quotation (RFQ)	Collecting competitive price quotes from minimum 3 providers	AI dataset annotation, penetration testing
Request for Proposal (RFP)	Seeking structured technical and commercial proposals for specialized work	Freelance AI/ML Engineer, external QA agency
Milestone-Based Contract	Fixed-scope contracts with phased payments tied to deliverable acceptance	Android developers, Backend engineer, DevOps
Short-Term Consultancy	Time-bound engagement for field operations and coordination	Field Coordinator (Pune)

3. Procurement Categories

3.1 Cloud Infrastructure & Tools

Item	Quantity	Cost (₹)	Procurement Method	Vendor Type
AWS EC2 + S3 + RDS + CloudFront	3-month plan	₹71,000	Direct Purchase	AWS India
Firebase (Blaze Plan)	3-month plan	₹12,000	Direct Purchase	Google Cloud
GPU Instance — AI Training	2-week burst	₹25,000	Direct Purchase	AWS India
Google Maps Platform API	Usage-based	₹7,000	Direct Purchase	Google Cloud
Firebase Test Lab Credits	Fixed credits	₹7,000	Direct Purchase	Google Cloud
Security Penetration Testing	1 engagement	₹30,000	RFQ (3 quotes)	Cybersecurity Firm
AI Image Dataset Annotation	5,000 images	₹40,000	RFP	Data Labeling Agency

3.2 Human Resources Procurement

Role	Duration	Cost (₹)	Contract Type	Sourcing
Senior Android Developer	3 months	₹1,80,000	Milestone Contract	Naukri / LinkedIn
Junior Android Developer	2 months	₹76,000	Fixed Contract	Freelance Platform
Backend Engineer (Node.js)	3 months	₹1,95,000	Milestone Contract	Developer Agency
AI/ML Engineer	2 months	₹1,40,000	RFP + Freelance Contract	Upwork / LinkedIn
UI/UX Designer	2 months	₹90,000	Freelance Contract	Behance / LinkedIn
QA Lead	2 months	₹1,00,000	Fixed Contract	QA Specialist Agency

Role	Duration	Cost (₹)	Contract Type	Sourcing
QA Tester (Junior)	1.5 months	₹45,000	Part-time Contract	Freelance Platform
DevOps Engineer	1 month	₹55,000	Short-term Contract	DevOps Specialist
Field Coordinator (Pune)	1 month	₹25,000	Short-term Consultancy	Local Agency

4. Vendor Selection Process

Step	Description
1. Market Research	Identify qualified developers, AI specialists, and cloud vendors on Upwork, LinkedIn, Naukri, and specialized referrals
2. Prequalification	Assess vendors on: domain experience (construction tech or equivalent), past client references, delivery track record, and pricing competitiveness
3. RFQ / RFP Issuance	Issue standardized Request for Quotation or Proposal to minimum 3 shortlisted vendors; specify scope, timeline, and deliverable criteria clearly
4. Technical Evaluation	Score proposals on: technical approach (40%), relevant experience (30%), cost (20%), timeline commitment (10%)
5. Negotiation & Contract Finalization	Negotiate payment milestones, IP ownership clauses, confidentiality agreements, and penalty terms for delays
6. Contract Awarding	Issue signed contracts; onboard selected vendors to project tools (Jira, Slack, Confluence)
7. Performance Monitoring	Track deliverable progress via Jira boards, weekly standups, and milestone-linked payment releases

5. Procurement Budget Summary

Category	Total Budget Allocated (₹)
Cloud Infrastructure & Tools (AWS, Firebase, Google Maps, Play Store)	₹1,18,000

Category	Total Budget Allocated (₹)
Software Tools & Licenses (Figma, Jira, Postman, Test Lab)	₹37,000
Human Resources (All Contract & Freelance Staff)	₹8,06,000
AI Dataset & Security Testing (Annotation + Pen Testing)	₹70,000
Field Operations, Contingency & Maintenance Reserve	₹1,19,000
Total Procurement Budget:	**₹12,50,000**

6. Risk Management in Procurement

Procurement Risk	Mitigation Strategy
Freelancer Dropout	Milestone contracts; 20% retention bonus on final delivery; backup candidate list maintained for all critical roles
Cloud Cost Overrun	AWS billing alerts at 80% of monthly budget; auto-scaling policies; weekly cost review
Vendor Delays	Penalty clauses for missed milestones; parallel evaluation for critical-path roles during onboarding
AI Dataset Quality Issues	Mandatory 10% sample review and quality audit before releasing full annotation payment
Tool Incompatibility	30-day trial or sandbox testing before committing to annual subscriptions
IP Ownership Dispute	Explicit IP assignment clauses in all contracts; work-for-hire terms for all deliverables

7. Procurement Timeline

Phase	Timeframe
Vendor Research & Prequalification	Week 1–2 (Feb 15–Feb 28, 2025)
RFQ / RFP Issuance & Proposal Collection	Week 2–3 (Feb 22–Mar 1, 2025)
Technical Evaluation, Scoring & Negotiation	Week 3–4 (Mar 1–Mar 7, 2025)

Phase	Timeframe
Contract Awarding & Tool Onboarding	Week 4–5 (Mar 8–Mar 14, 2025)
Development Sprint 1 Kickoff	Week 5 (Mar 15, 2025)
Mid-Project Review & Milestone Payment Release	Week 7–8 (Mar 29–Apr 5, 2025)
Final Deliverables, Deployment & Close-Out Payments	Post Month 3 (May 2025)

PROJECT CHANGE CONTROL

All changes to project scope, schedule, or budget are formally managed through this Change Control process. No change is implemented without completing a Change Request Form and receiving written approval from the authorized approver.

Change Request Template

CHANGE REQUEST FORM	
Change Request ID	CR-[001]
Date Submitted	[DD/MM/YYYY]
Submitted By	[Name, Role]
Change Title	[Brief description of proposed change]
Change Category	Scope / Schedule / Budget / Quality / Risk [circle one]
Description of Change	[Detailed description of what is being changed and why]
Reason / Justification	[Business reason, technical necessity, or user feedback driving the change]
Impact on Scope	[Additions or removals to project deliverables]
Impact on Schedule	[Number of days added or removed; affected activities]
Impact on Budget	[Additional cost or cost savings; revised budget if applicable]
Impact on Quality / Risk	[New risks introduced; quality metrics affected]
Recommended Action	Approve / Reject / Defer / Escalate
Approved By	[Project Manager / Sponsor — Name & Signature]
Approval Date	[DD/MM/YYYY]
Implementation Notes	[Sprint or phase where change will be incorporated]

Implemented Change Log

CR ID	Change Title	Category	Description	Status	Approved By
CR-001	Offline-First Architecture Enhancement	Scope / Technical	Revised original online-only architecture to incorporate offline data queuing after field visits to Pune construction sites revealed severe connectivity limitations for target workers.	Approved	Project Manager
CR-002	AI Tagging UI Feedback Loop	Scope / UX	Added icon-tag combinations for AI results, one-tap 'dispute tag' feature, and confidence score bar display based on internal QA review and mock UAT feedback.	Approved	Project Manager
CR-003	QR Verification Flow Simplification	Scope / UX	Reduced 6-step supervisor approval form to 2-tap workflow; added pre-population of work details and bulk verification mode to reduce friction and improve supervisor participation.	Approved	Project Manager + HoE

All three changes were assessed for budget and schedule impact, integrated within existing sprint cycles, and resulted in no change to the overall project budget (₹12,50,000) or completion date (May 15, 2025).

PROJECT MONITORING AND CONTROLLING

Objective: To ensure the Skill Passport platform remains aligned with its approved scope, schedule, and budget baselines while proactively managing risks and maintaining quality standards throughout the development and pilot phases.

1. Performance Tracking and Schedule Control

- Milestone Reviews: Sprint reviews at the end of each biweekly sprint assess feature completion status against the WBS activity schedule.
- Progress Reports: Biweekly status reports submitted by each team lead (Android, Backend, AI/ML, QA) to the Project Manager using a standardized template.
- Task Tracking: Jira sprint boards track all tasks across To-Do, In-Progress, In-Review, and Done stages with real-time visibility.
- Gantt Chart Maintenance: Project master schedule updated weekly to reflect actual progress and flag delayed activities.
- EVM Tracking: Monthly Earned Value Analysis calculating CPI and SPI to assess budget and schedule performance.

2. Quality Assurance and Validation

- Weekly automated regression test suites triggered by CI/CD pipeline on every code merge to the development branch.
- AI Tagging Accuracy Reports generated after each model iteration; escalation triggered if accuracy falls below 78%.
- Backend API Monitoring via AWS CloudWatch tracking response times, error rates, and uptime against the 99.5% SLA.
- UAT Findings documented post-session; task completion rates and worker satisfaction scores reviewed by Project Manager.

3. Risk Monitoring and Mitigation

- Risk Register updated weekly; new risks logged with probability, impact, and response assignment.
- Low worker adoption: Field Coordinator deployed at pilot sites for hands-on app guidance and assisted onboarding.
- AI accuracy drops: Automatic retraining pipeline triggered when monthly audit falls below 78% threshold.
- Supervisor disengagement: Weekly check-in calls with participating contractor firms to monitor QR verification participation rates.

4. Cost and Resource Control

- Monthly Budget Burn Rate Review: Actual vs. planned expenditure reconciled against each WBS cost category by the Project Manager.
- AWS Billing Alerts set at 80% of monthly cloud budget threshold; usage reviewed and optimized weekly by DevOps Engineer.
- Freelancer Milestone Tracking: Payment releases tied to deliverable acceptance by the QA Lead; withheld for incomplete or non-compliant deliverables.
- Cost Performance Index (CPI) tracked monthly; any CPI < 0.9 triggers a formal Cost Review with the project sponsor.

5. Communication and Stakeholder Engagement

- Daily Standups: 15-minute all-team sync covering progress, blockers, and daily priorities.
- Biweekly Sprint Review: PM, Head of Engineering, and QA Lead assess sprint delivery and plan next sprint commitments.
- Monthly Stakeholder Briefings: Project summary presented to sponsors and partner contractors covering KPI actuals, adoption data, risk status, and upcoming milestones.
- Issue and Escalation Log maintained in Confluence; all escalations documented with resolution actions and timelines.

6. Change Management

- All proposed changes to scope, schedule, or budget are submitted via a formal Change Request Form (see Project Change Control section).
- Project Manager reviews and assesses impact within 48 hours; changes exceeding ₹50,000 or 3 days of schedule impact require sponsor approval.
- Approved changes are logged in the Change Register, incorporated into the updated project baseline, and communicated to all affected team members.

7. Final Control Checkpoints

- Pre-Deployment Readiness Review: Full regression test suite executed; AI tagging accuracy ≥ 80%; QR verification flow tested under simulated field conditions; security pen test report sign-off confirmed.
- UAT Sign-Off: Final UAT completed with minimum 20 construction workers at Pune pilot site before Google Play Store submission.
- Go / No-Go Decision: Formal Go/No-Go meeting with Project Manager and Head of Engineering before app store publication.

- Post-Deployment Monitoring: 30-day active monitoring period tracking worker onboarding rate, QR verification rate, contractor search engagement, API error rates, and system uptime.

Updated Risk Register — Monitoring Status

ID	Risk	Risk Response Implemented	Responsible	Current Status	Risk Level
R1	Low Digital Literacy	Icon-based UI deployed; regional language support added; field coordinator providing on-site guidance	UI/UX Lead, Field Coordinator	Onboarding Support Active — drop-off rate monitored	Medium
R2	Supervisor Reluctance	2-tap QR flow live; bulk verification mode added; contractor firm MoU signed	PM, HoE	Participation Rate Improving — 68% completion rate	Medium
R3	Poor Image Quality	In-app quality checker and framing overlay deployed; low-res rejection active	Android Dev, QA Lead	Quality Filter Active — rejection rate at 12%	Low
R4	Network Failure	Offline-first queue live; background sync tested; 48-hour sync buffer implemented	DevOps, Backend	Offline Mode Active — no data loss incidents reported	Medium
R5	Data Privacy Breach	AES-256 + TLS 1.3 active; pen test completed; DPDP compliance confirmed	Backend, Compliance	Compliant — Zero vulnerabilities in pen test report	Low

ID	Risk	Risk Response Implemented	Responsible	Current Status	Risk Level
R10	Freelancer Dropout	Milestone contracts active; 20% retention bonus applied; backup candidate pre-identified	PM	Stable — All contractors meeting milestone targets	Low

PROJECT SUMMARY

Performance Indicator	Planned / Target Value
Total Estimated Effort	289 Person-Days across all team members
Total Duration	89 Calendar Days (Feb 15 – May 15, 2025)
Critical Path Sequential Sum	90 Days: 1.1 → 1.2 → 1.3 → 1.5 → 1.6
Project Start Date	February 15, 2025
Project End Date	May 15, 2025
Total Project Budget	₹12,50,000 (Twelve Lakhs Fifty Thousand Rupees)
Platform Type	Android Mobile Application — Worker, Supervisor, and Contractor Modules
Target User Base — Workers	200 verified construction worker profiles (Pilot)
Target User Base — Contractors	50 registered contractor accounts (Pilot)
AI Skill Tagging Accuracy Target	≥ 80% validated through QA audit
QR Verification Turnaround Target	≤ 24 hours from scan to verified badge
Platform Uptime Target	≥ 99.5% during 90-day post-launch monitoring
Worker Satisfaction Score Target	≥ 80% positive (post-onboarding in-app survey)
API Response Time Target	≤ 2 seconds under standard load
App Load Time Target (3G)	≤ 5 seconds from launch to home screen
App APK Size Target	≤ 25 MB
Deployment Platform	Google Play Store — India (Pune Pilot)
Number of Identified Risks	10 (4 High Probability, 5 Medium Probability, 1 Low Probability)
Number of Approved Change Requests	3 (CR-001, CR-002, CR-003)
Number of WBS Work Packages	29 across 6 major phases

The Construction Worker Skill Passport with Project Portfolio represents a transformative and long-overdue intervention in India's informal construction labour market. By combining mobile photo documentation, QR-based peer verification, and AI-powered skill tagging, the platform creates a portable, contractor-agnostic credential system that has never previously existed for this workforce of over 51 million individuals.

The 89-day development and pilot cycle, executed within a ₹12,50,000 budget, demonstrates the technical and financial viability of building scalable social infrastructure on lean, purpose-built technology. The platform's offline-first architecture and visual, icon-driven UI address the two most significant adoption barriers — connectivity and digital literacy — making it practically deployable in real construction site conditions.

Upon successful completion of the pilot phase, the project is positioned for expansion to additional cities, integration with government contractor registration portals (e-SHRAM), and potential formal linkage with skill certification bodies such as the National Skill Development Corporation (NSDC) and the Construction Sector Skill Development Council (CSSDC).

Project Closure Signatures

Role	Name & Signature
Project Manager, SkillBridge Technologies Pvt. Ltd.	[Your Name Here] ______________________________ Date: May 15, 2025
Head of Engineering	Arjun Sharma ______________________________ Date: May 15, 2025
QA & Testing Lead	Rahul Mehta ______________________________ Date: May 15, 2025

APPENDIX A — AUDIT FINDINGS AND CORRECTIONS

This appendix documents all errors identified during the professional audit of the original workbook and every correction applied across versions 2.0 and 3.0. Version 3.0 introduces 15 additional corrections beyond the original 23, totalling 38 documented corrections.

Version 2.0 Corrections (23 issues resolved)

#	Error Found	Why It Was Wrong (PMBOK Reference)	Correction Applied
1	Missing Cost Management Plan	PMBOK 6th Ed. Process 7.1 requires a Cost Management Plan as a mandatory subsidiary plan	Added dedicated Cost Management Plan with estimation method, control thresholds, EVM approach, and reserve strategy
2	Missing RACI Matrix	PMBOK Resource Management requires role-responsibility mapping; without RACI, accountability is undefined	Added complete RACI Matrix (12 activities × 7 roles) under Human Resource Management Plan
3	Missing Change Request Template	PMBOK Process 4.6 (Perform Integrated Change Control) requires a formal change request form	Added standardized Change Request Form template with all required PMBOK fields
4	Missing Quality Reporting Plan	PMBOK Process 8.1 requires a documented quality reporting approach	Added Quality Reporting Plan table with 6 report types, frequencies, and contents
5	Schedule overlap: Design (1.2) and Development (1.3) both started Mar 1	Development cannot start before Design is complete — violates Finish-to-Start dependency	Corrected: Design ends Mar 14; Development starts Mar 15, respecting FS dependency
6	Activity 1.4 (AI Dev) had no dependency justification	No explanation for why AI development could start before core development was complete	Corrected: AI Dev runs parallel to Core Dev (Mar 15–Mar 28, SS dependency) with float analysis

#	Error Found	Why It Was Wrong (PMBOK Reference)	Correction Applied
7	Activity 1.6 duration 27 days but dates didn't reconcile	Apr 19 + 27 days = May 16, not May 15; dates were inconsistent	Corrected deployment phase Apr 19–May 15 (27 days); App Published milestone on Apr 25
8	Budget double-counting in 'Core Development & Other Costs'	Same costs listed twice inflated true allocation; manpower was understated	Eliminated duplicate category; reconciled all four cost lines to ₹12,50,000
9	Manpower total wrong; QA Tester missing	QA Tester role referenced in activities but absent from manpower table	Added QA Tester (₹45,000); corrected Junior Dev to ₹38,000/month; new total ₹8,06,000
10	Stakeholder register listed only 4 stakeholders	All project participants must appear in register (PMBOK 13.1)	Expanded register to 10 stakeholders including DevOps, Field Coordinator, Workers, Contractors
11	Power-Interest grid mapped only 4 of 10 stakeholders	Incomplete stakeholder analysis	Updated grid to show all 10 stakeholders across 4 quadrants
12	WBS lacked Level 1 root node	PMBOK 5.4 requires WBS to start from Level 1 (the project itself)	Added '1.0 Construction Worker Skill Passport Platform' as Level 1 root node
13	WBS missing 1.3.5 Worker Rating System	Every feature in scope must appear in WBS	Added 1.3.5 Worker Rating & Endorsement System as a work package
14	WBS missing 1.5.5 Security Penetration Testing	Security testing is a distinct work package with separate contractor and budget	Added 1.5.5 Security Penetration Testing under 1.5
15	Resource estimation missing QA Tester	QA Tester listed in team breakdown but not in resource estimation	Added QA Tester (Junior) to activity resource estimation table

#	Error Found	Why It Was Wrong (PMBOK Reference)	Correction Applied
16	Risk matrix had empty cells without explanation	A complete risk matrix should show '—' for empty cells, not leave them blank	Rebuilt risk matrix with all 10 risks correctly placed; empty cells explicitly shown
17	Procurement budget categories inconsistent with Cost Estimation	Category names differed between Cost Estimation and Procurement Plan	Aligned procurement budget categories to match Cost Estimation categories exactly
18	No Change Request Form — only a changelog	PMBOK Process 4.6 requires a formal change request mechanism	Added formal CR Form template before the implemented change log
19	Monitoring section formatting issue	Sub-bullets promoted to main bullets made accountability unclear	Corrected: risk monitoring items properly formatted as subordinated bullets
20	Project Summary missing total effort in person-days	Summary should enable quick assessment of resource intensity	Added 289 person-days total effort figure
21	No critical path calculation	PMBOK 6.3 (Develop Schedule) requires critical path identification	Added Critical Path section: 1.1→1.2→1.3→1.5→1.6 = 90 days; 1.4 has 7 days float
22	Activity dependency table missing	PMBOK requires activity sequencing documentation	Added Activity Dependency table showing FS/SS relationships for all 6 phases
23	Communication plan had no tools reference	Tools are part of Communications Management Plan (PMBOK 10.1)	Added Communications Tools table listing Slack, Jira, Confluence, Meet, CloudWatch, Google Workspace

Version 3.0 Corrections (15 additional issues resolved)

#	Error Found	Why It Was Wrong	Correction Applied
24	Appendix A appeared at the top of the document before main content	Appendices must follow all main sections in a professional document	Moved Appendix A to end of document after Project Summary and Closure Signatures
25	Inconsistent placeholder names: '[Your Name]' vs '[Your Name Here]'	All placeholder instances should use the same standardized format throughout	All placeholders standardized to '[Your Name Here]' throughout the document
26	Duration inconsistency: Phase Summary total row showed '89 Days' while activity table total row showed '90'	These two values refer to different measures (calendar span vs. sequential sum) but were presented without explanation, creating confusion	TOTAL row now shows '90 days (critical path sequential sum)' with a note that the calendar span (Feb 15–May 15) = 89 days; both values explained
27	Activity 1.4.2 resource mismatch: listed 'AI/ML Engineer (1)' = 5 PD but stated 10 PD	5 days × 1 person = 5 PD, not 10 PD as stated in the table	Resource corrected to 'AI/ML Engineer (1), Backend Engineer (1)' = 5 × 2 = 10 PD ✓
28	Activity 1.4.5 resource mismatch: listed 'Senior Android Developer (1)' = 4 PD but stated 8 PD	4 days × 1 person = 4 PD, not 8 PD as stated	Resource corrected to 'Senior Android Developer (1), Junior Android Developer (1)' = 4 × 2 = 8 PD ✓
29	Activity 1.5.5 resource ambiguous: 'External Security	'External Security Firm' with no headcount makes the 6 PD calculation impossible to verify	Clarified to 'External Security Firm (2-person team)' = 3 × 2 = 6 PD ✓

#	Error Found	Why It Was Wrong	Correction Applied
	Firm' with 3 days and 6 PD		
30	Activity 1.5.6 resource mismatch: 'QA Lead(1), Field Coordinator(1), 10 Workers' listed, 5 days, stated 20 PD	10 construction workers are UAT participants (end users), not project staff resources. Including them would give 60 PD, not 20 PD	Resource list corrected to 'QA Lead (1), Field Coordinator (1), UAT Facilitators (2)' = 5 × 4 = 20 PD ✓
31	Risk count in Project Summary stated '12 (4 High, 6 Medium, 2 Low)' but risk register lists exactly 10 risks	The breakdown does not match the risk register (R1–R10). Actual count: 4 High, 5 Medium, 1 Low	Project Summary corrected to: '10 risks (4 High Probability, 5 Medium Probability, 1 Low Probability)'
32	Procurement Section 4 (Vendor Selection Process) was missing — document jumped from Section 3 to Section 5	Skipping a numbered section creates a documentation gap and gives the impression of an incomplete plan	Section 4 (Vendor Selection Process) restored with 7-step evaluation framework
33	WBS items contained '[ADDED — was missing from original WBS]' inline audit annotation labels	Audit annotations are internal notes, not appropriate content in a final submission document	All WBS audit annotation labels removed; clean hierarchical WBS retained
34	Manpower table entry 23 showed 'QA Tester (Junior) [ADDED]' with the audit label inline	Same issue as above — audit labels must not appear in final document body	'[ADDED]' label removed from manpower table; entry retained with clean formatting

#	Error Found	Why It Was Wrong	Correction Applied
35	'CORRECTED NOTE:' paragraph appeared in the Activity Dependency section	This was an audit/reviewer comment embedded in the document body, inappropriate for final submission	Removed the inline correction note; clean dependency table retained with proper context
36	Phase 1.4 in phase summary labelled '14 (parallel)' but footer note said '14 days' without reconciling the 89/90 discrepancy	The dual labelling without explanation created reader confusion about whether 89 or 90 was authoritative	Label standardized to '14 (parallel with 1.3)' and the TOTAL row now fully explains both measures
37	Budget Verification note was informal inline text rather than a clearly labelled line	In a professional workbook, verification statements should be clearly set off from narrative text	Budget verification statement retained but formatted consistently after the grand total table
38	Procurement section missing Sections 6 (Risk Management in Procurement) and 7 (Procurement Timeline) which were present in the original but had been relocated	Complete procurement plan requires risk and timeline subsections per PMBOK 12.1	Sections 6 and 7 restored to Procurement Management Plan in correct order